



WEEK 5 TASK

IMDB Reviews Classification



AUGUST 19, 2025

DEN

STU-DS-251-232

Contents

Objective	1
Approach.....	1
Results.....	2
Conclusion.....	3

IMDB Movie Reviews Sentiment Analysis

Author: MUHAMMAD KAIF UR REHMAN

Course: Data Science — Week 05

Mentor: Ali Mohiuddin Khan

Objective

The goal of this project is to build a **binary text classification model** that predicts whether an IMDB movie review is **positive** or **negative**. We apply **Natural Language Processing (NLP)** techniques and train a **Naive Bayes classifier** on the IMDB dataset.

Approach

1. Data Loading & EDA

- Dataset: **50,000 IMDB reviews** labeled as positive or negative.
- Removed **418 duplicate rows** and verified there were no missing values.
- Sentiment distribution was **balanced** (nearly equal positive and negative reviews).

2. Preprocessing

- Encoded positive → 1 and negative → 0.
- Cleaned reviews by:
 - Removing HTML tags and punctuation.
 - Lowercasing all text.

- Removing stopwords (e.g., “the”, “and”).
- Created a new column `clean_review` with the processed text.

3. Feature Extraction

- Used **TF-IDF Vectorizer** with a vocabulary size of **5000 features**.
- Converted the reviews into sparse numerical vectors for model training.

4. Model Training

- Trained a **Multinomial Naive Bayes** classifier on 80% of the data.
- Evaluated on the remaining 20% test set.

5. Evaluation Metrics

- **Accuracy:** 85.2%
- **Precision / Recall / F1-score:** Balanced across both classes (positive and negative).
- **Confusion Matrix** showed nearly equal performance in detecting both sentiments.

6. Interpretability

- Extracted most indicative words:
 - **Positive words:** great, excellent, love, amazing, wonderful, etc.
 - **Negative words:** boring, worst, bad, waste, terrible, etc.

7. Saving the Model

- Exported trained **Naive Bayes model** and **TF-IDF vectorizer** using pickle for future use.

Results

- The model achieved **85% accuracy**, showing strong baseline performance.
- Precision and recall are nearly identical across classes, which confirms that the model is not biased towards one sentiment.
- The analysis demonstrates that even a relatively simple model like **Naive Bayes** with TF-IDF features can perform effectively on real-world text classification tasks.

Conclusion

This project successfully built a sentiment analysis system on IMDB reviews.

- The pipeline is **efficient, interpretable, and reliable**.
- It can serve as a **baseline model** for more advanced techniques such as logistic regression, SVM, or deep learning (LSTM/Transformers).
- Future improvements could involve **n-grams, hyperparameter tuning, and advanced embeddings (Word2Vec, BERT)** to further boost accuracy.